Constrains in PCB Editor

This document outlines the constraints you will need to set in PCB Editor, so that the electronics workshop will be able to manufacture your board properly. The constrains menus defines limits on certain dimensions, like the track widths, or for the permitted spacing between tracks, pads and vias.

The below constraints will work for most PCBs with through holes or simple surface mount devices (SMDs). For more advanced work (SMD devices with a 0.5mm pitch and 0.2mm gap) these may have to be revised. Please contact Andrew Phillips if you are using non-standard components (andrew.phillips@glasgow.ac.uk).

Inside the PCB Editor software:

From the menu system select: SETUP → CONSTRAINTS

WORKSHEET > PHYSICAL
Physical Constraint set > All Layers.

Line width min = 0.5mm (20mil)

Line width Max = 10mm (400mil)
(Adjust depending on current requirements for Tracks)

Neck min width = 0.2mm (8mil)
(For when you need to squeeze a track between 2 pads for a short distance, use with caution.)

Max Length = 20mm (800mil)
(This can be tweaked if using large quad flat packs which require a large amount of close tracks)

Note:

- Values can also be set per Net ("Net→All layers"). E.g. VCC & GND can be set to wider tracks)
- Design with as large tracks and gaps as possible, use 'neck' when routing (thinner region) to connect to surface mount components with small pin pitch. You can do this by right clicking when routing and selecting 'neck'.

WORKSHEET > SPACING
Spacing Constraint Set > All Layers

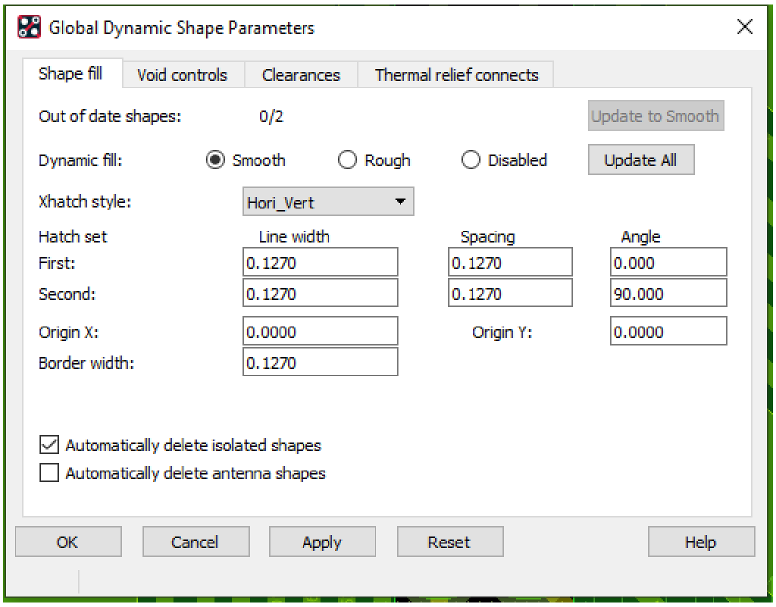
Line to all = 0.5mm (20mil)
Thru pin to all = 0.5mm (20mil)

On the **"SMD pin to"** box, double click, and it will show more options (rather than just 'to all').
SMD pin to line = 0.2mm (8mil)
SMD pin to SMD pin = 0.2mm (8mil)

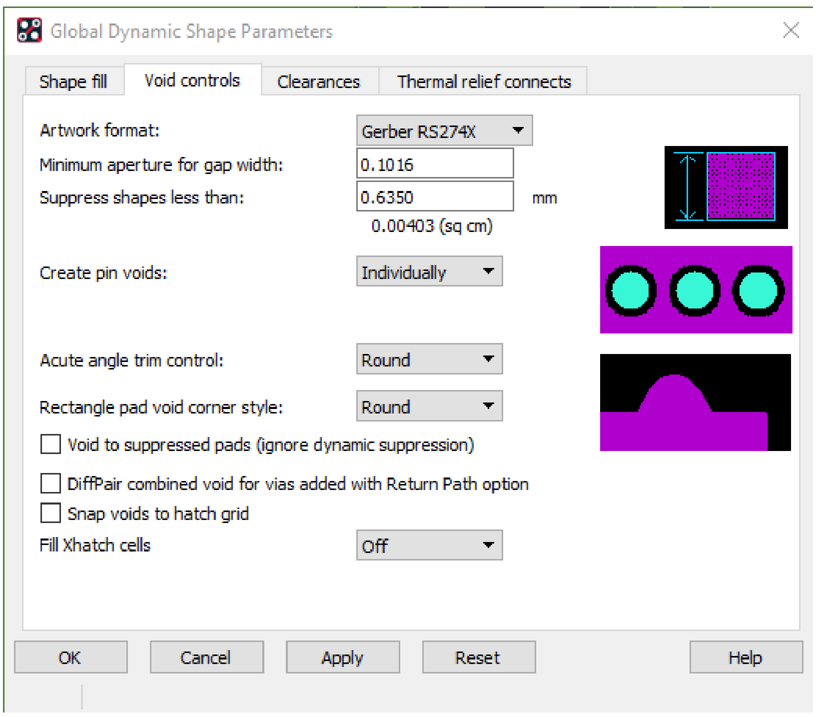
Thru Via to All = 0.5mm (20mil)
Shape to all = 0.5mm (20mil)

Close constrains window

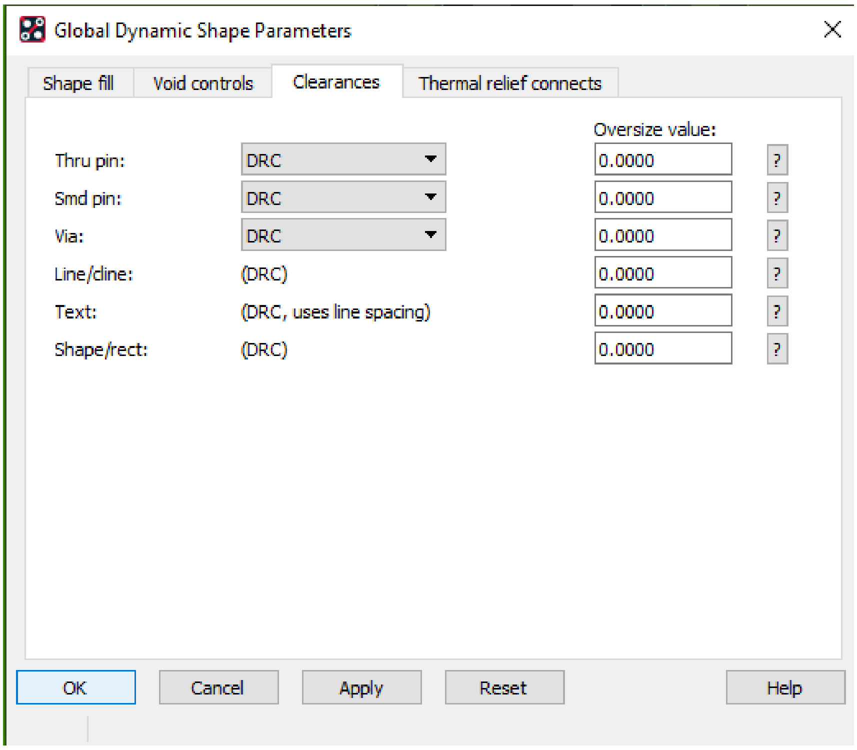
From the menu system select: SHAPE > GLOBAL DYNAMIC SHAPE PARAMETERS



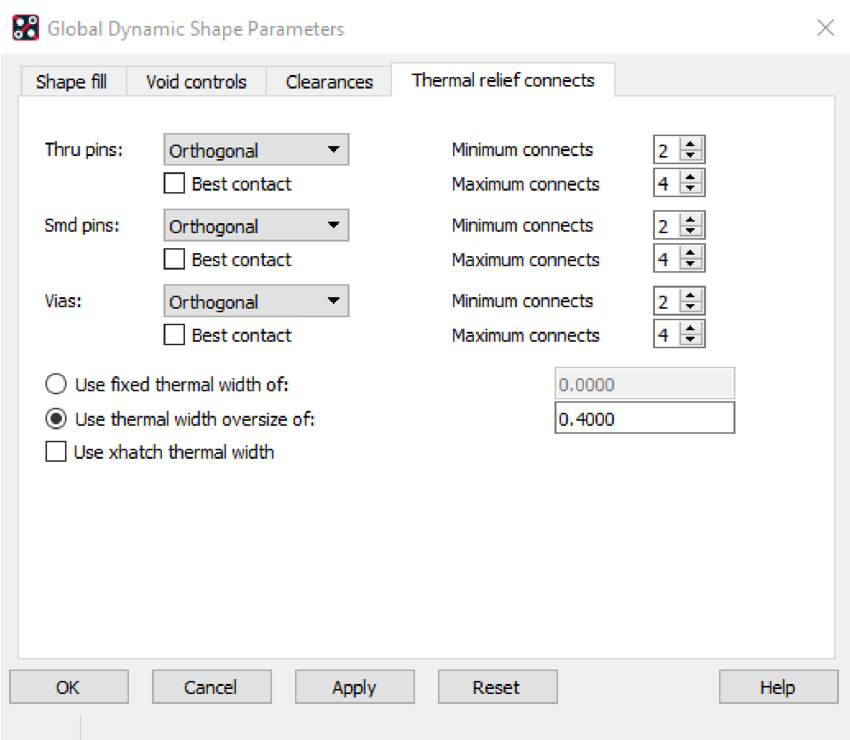
- Set Dynamic fill to smooth,
- Check Update All
- Leave other settings as default.
- **NOTE: The above dimension are in mm. 0.1270 mm = 5 mils**



- Ensure artwork format is set to Gerber RS274X



- Leave Clearances set to DRC



- Adjust thermal reliefs as above
- Avoid best contact as it can make the pads merge with the fill, making it difficult to solder.